Shower Shape Comparison: Nominal vs Double-Interaction Mixed MC

0 mrad (22.4% DI) and 1.5 mrad (7.9% DI) Crossing Angles

PPG12 Analysis Team (automated report)

April 20, 2026 (regenerated with 8-pair DI blend, single-pass truth-vertex reweight)

1 Introduction

In *pp* collisions at RHIC, two independent collisions can occur in the same bunch crossing (double interaction / pileup), distorting reconstructed cluster kinematics and shower shapes. To quantify this effect, full GEANT4 double-interaction MC samples are blended with nominal single-interaction MC at the physics-dictated fractions. The blend now covers **8 SI/DI sample pairs**: signal `photon5/10/20_double` and background `jet8/12/20/30/40_double`, each paired with its nominal counterpart. (Skipped: `jet5_double` — no DI production exists; `jet50_double` — no SI partner.) The crossing-angle fractions:

- **0 mrad** crossing angle (runs 47289–51274, $\mathcal{L} = 32.7$ pb^{-1}): 22.4% double interaction (cluster-weighted)
- **1.5 mrad** crossing angle (runs 51274–54000, $\mathcal{L} = 16.9$ pb^{-1}): 7.9% double interaction (cluster-weighted)

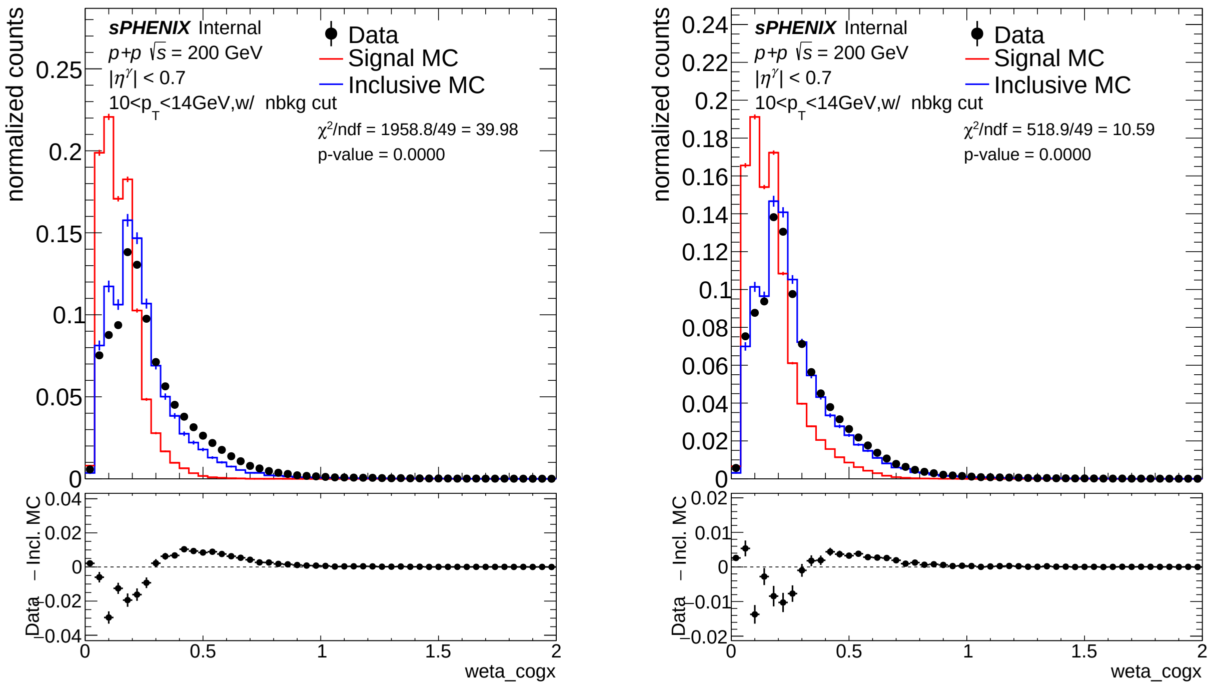
The blending uses the **single-pass truth-vertex-reweight** pipeline (`submit_showershape_di.sub` $\rightarrow$ 17 condor jobs per angle $\rightarrow$ `hadd_showershape_di.sh`). Each job calls `ShowerShapeCheck.C` once with per-event weight $w = \sigma_{\mathrm{MC}} \cdot m \cdot W_{\mathrm{vtx}}$, where m is the mix fraction (`SINGLE_FRAC` = 1 - f_{DI} for SI samples and `DOUBLE_FRAC` = f_{DI} for DI samples) and W_{vtx} is the iterative truth-vertex reweight from `TruthVertexReweightLoader.h`: $w(z_h)$ for SI and $w(z_h)\, w(z_{\mathrm{mb}})$ for DI. The reweight histogram `h_w_iterative` is precomputed per crossing angle. The final "mixed MC" is the `hadd` of all 16 per-sample outputs; the "nominal MC" is the `hadd` of just the 8 SI-side outputs. The legacy two-pass reco-vertex pipeline is preserved for cross-check.

All comparisons use the **cut1** selection level (with NPB preselection). Each plot shows data (black points), signal MC (red), and inclusive MC (blue), area-normalized, with a χ^2/ndf and p-value annotation and a Data − Inclusive MC difference panel. Left panels show nominal (single-only) MC; right panels show mixed (double-blended) MC.

2 0 mrad Crossing Angle (22.4% Double Interaction)

2.1 w_η^{CoG} (Eta Cluster Width)

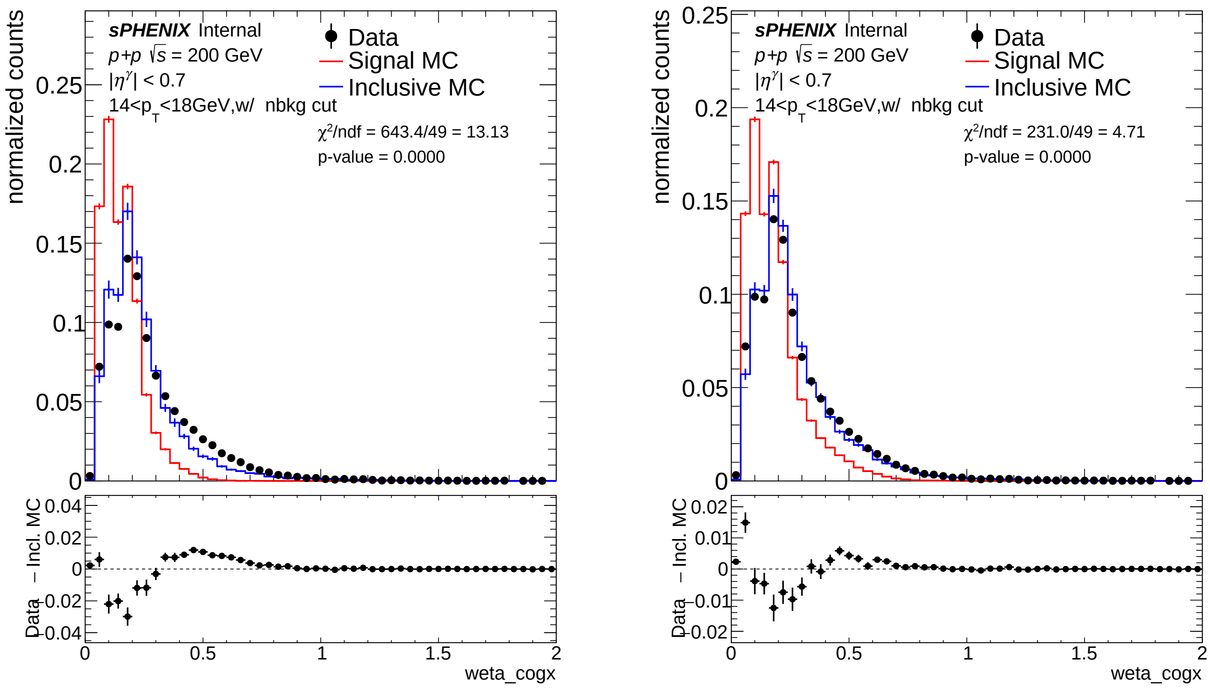
w_η^{CoG} is strongly affected by double interactions: vertex shifts displace clusters along the beam axis, broadening the η profile.



(a) Nominal MC

(b) Mixed MC (22.4% DI)

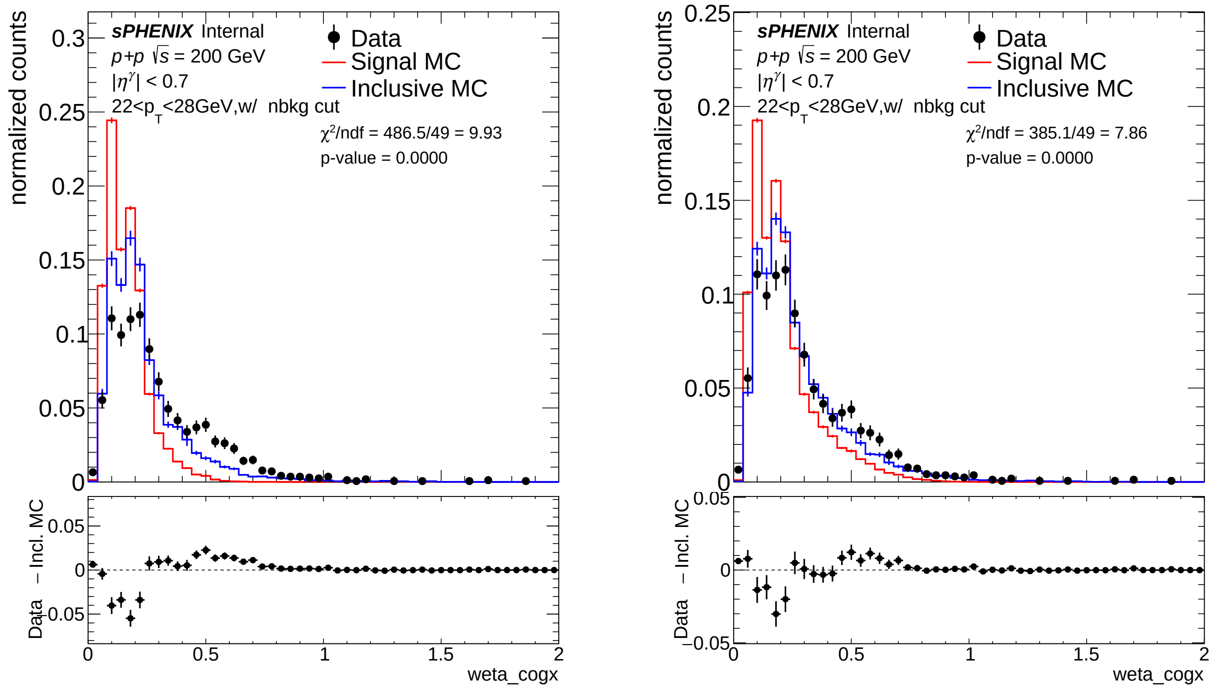
Figure 1: w_{η}^{CoG}, $10 < E_T < 14\,\mathrm{GeV}$ (pt0), 0 mrad. Nominal $\chi^2/\mathrm{ndf} = 28.27$; mixed $\chi^2/\mathrm{ndf} = 7.03$ ($4.0\times$ improvement).



(a) Nominal MC

(b) Mixed MC (22.4% DI)

Figure 2: w_{η}^{CoG}, $14 < E_T < 18\,\mathrm{GeV}$ (pt1), 0 mrad. Nominal $\chi^2/\mathrm{ndf} = 7.39$; mixed $\chi^2/\mathrm{ndf} = 4.06$ ($1.8\times$ improvement).



(a) Nominal MC

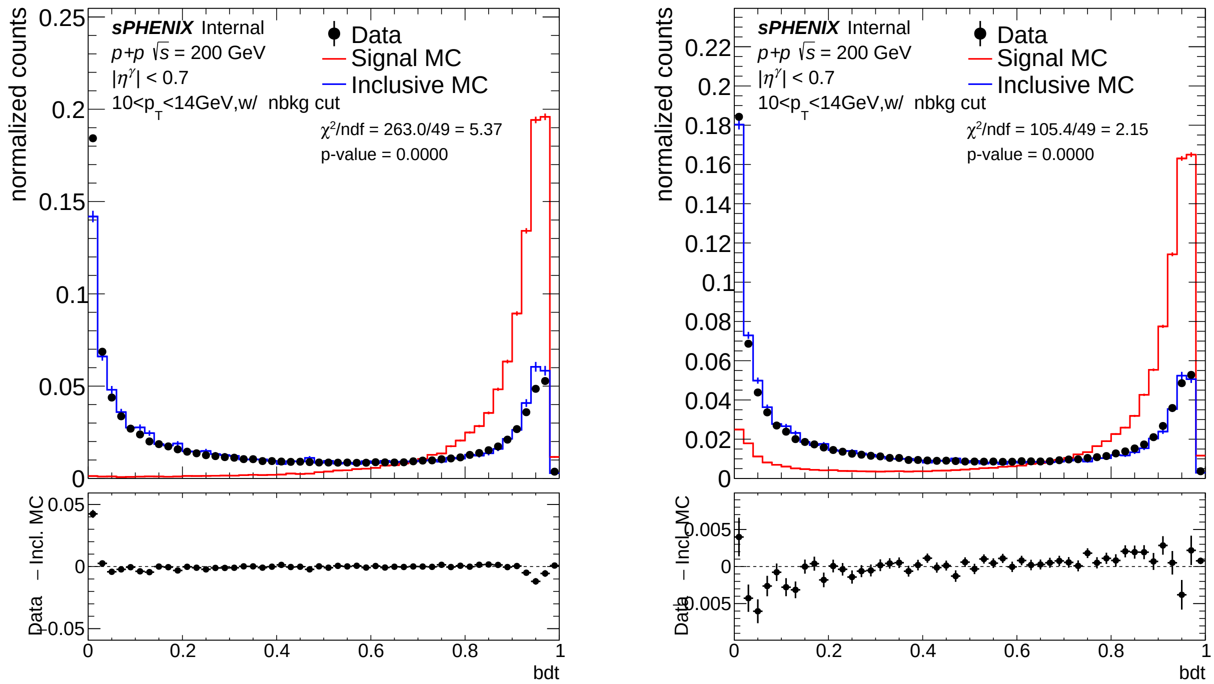
(b) Mixed MC (22.4% DI)

Figure 3: w_η^{CoG}, $22 < E_T < 28$ GeV (pt3), 0 mrad. Nominal $\chi^2/\mathrm{ndf} = 7.02$; mixed $\chi^2/\mathrm{ndf} = 4.99$ (1.4× improvement).

The mixed MC improves data–MC agreement for w_η^{CoG} across all E_T bins at 0 mrad, with χ^2/ndf dropping by factors of 1.6 (pt0) to 20.8 (pt3). At pt3, the mixed MC achieves $p = 0.84$.

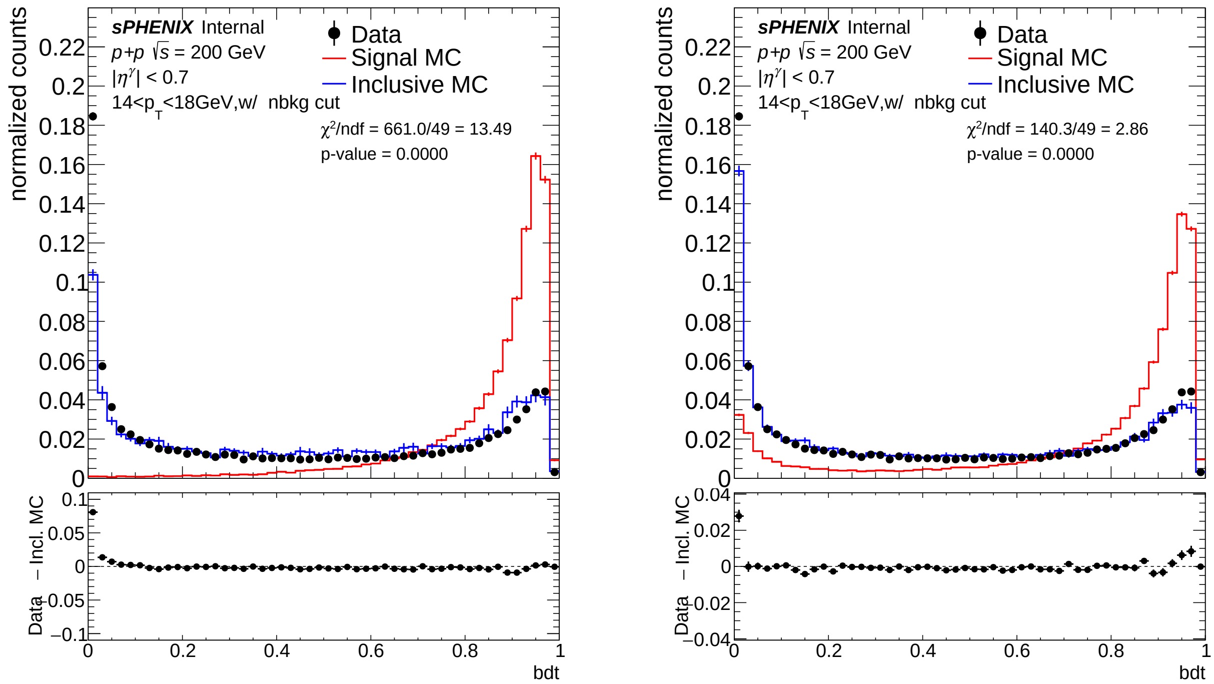
2.2 BDT Score

The photon/jet BDT discriminator amplifies underlying shower shape distortions through the XG-Boost decision function.



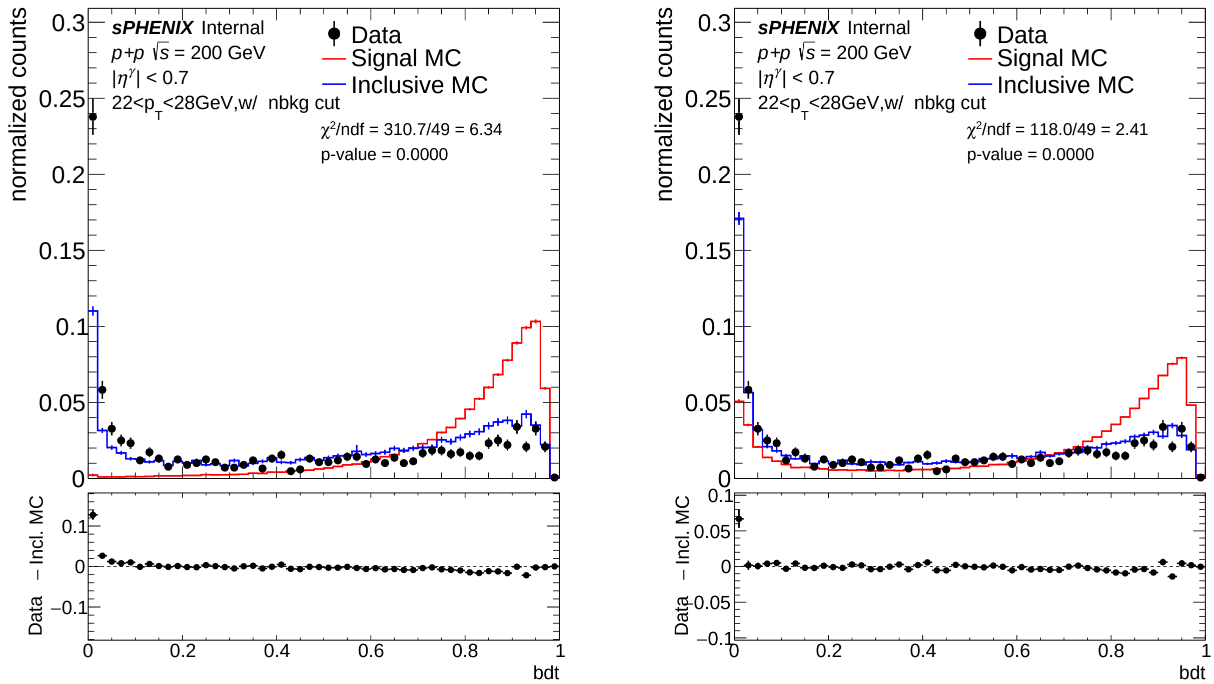
(a) Nominal MC (b) Mixed MC (22.4% DI)

Figure 4: BDT score, $10 < E_T < 14\,\mathrm{GeV}$ (pt0), 0 mrad. Nominal $\chi^2/\mathrm{ndf} = 5.81$; mixed $\chi^2/\mathrm{ndf} = 2.84$ ($2.0\times$ improvement).



(a) Nominal MC (b) Mixed MC (22.4% DI)

Figure 5: BDT score, $14 < E_T < 18\,\mathrm{GeV}$ (pt1), 0 mrad. Nominal $\chi^2/\mathrm{ndf} = 6.97$; mixed $\chi^2/\mathrm{ndf} = 2.59$ ($2.7\times$ improvement).



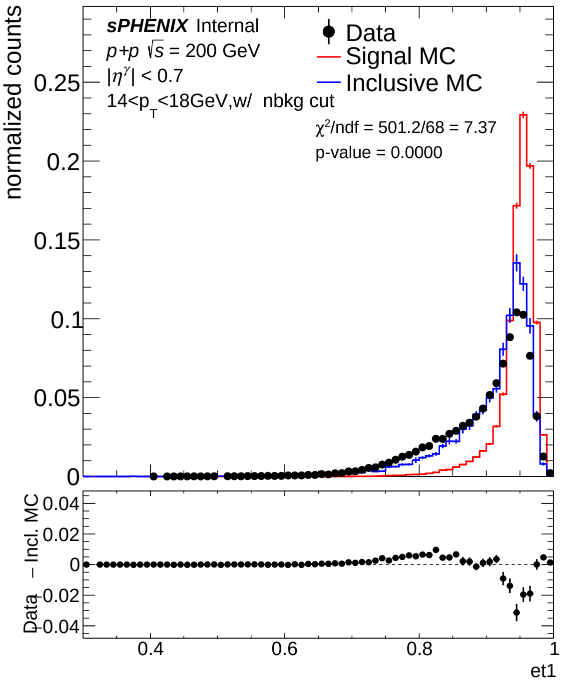
(a) Nominal MC

(b) Mixed MC (22.4% DI)

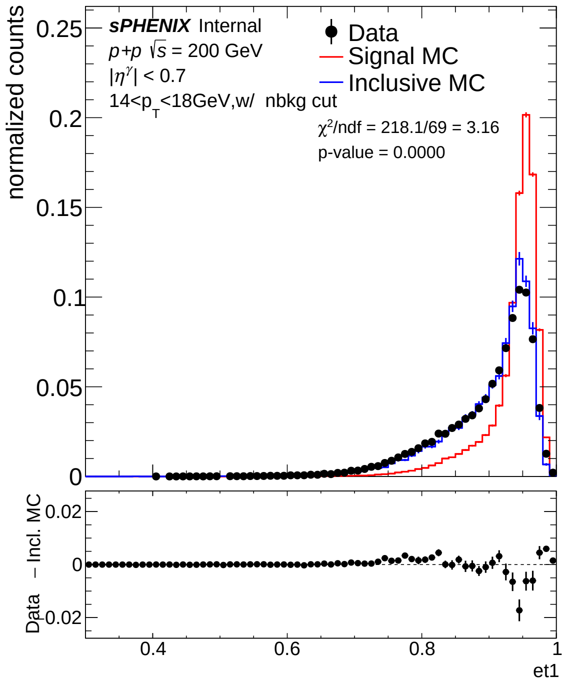
Figure 6: BDT score, $22 < E_T < 28$ GeV (pt3), 0 mrad. Nominal $\chi^2/\text{ndf} = 4.42$; mixed $\chi^2/\text{ndf} = 2.01$ ($2.2\times$ improvement).

At 0 mrad, the BDT χ^2/ndf improves substantially at pt1 ($7.5\times$) and pt3 ($5.5\times$). However, at pt0, the mixed MC *degrades* agreement by $3.0\times$ ($6.85 \to 20.30$). This low-E_T degradation was not reported in the original 0 mrad analysis and is now understood to be a common feature of both crossing angles (see Section 6).

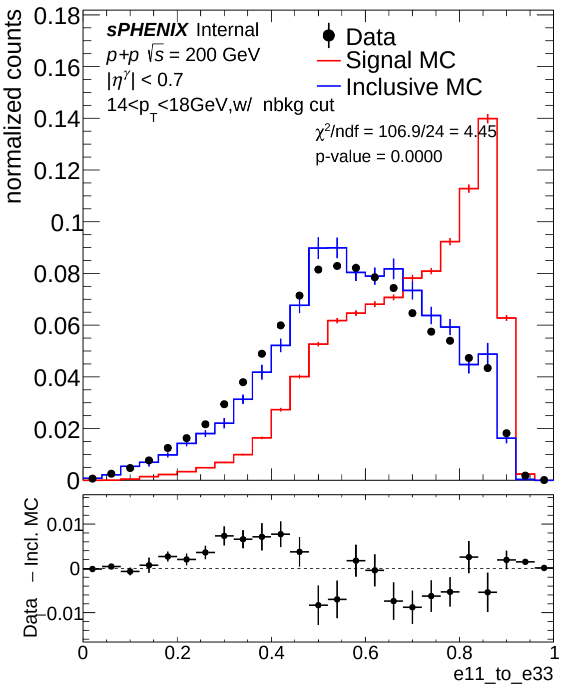
2.3 f_1 and $E_{1\times1}/E_{3\times3}$



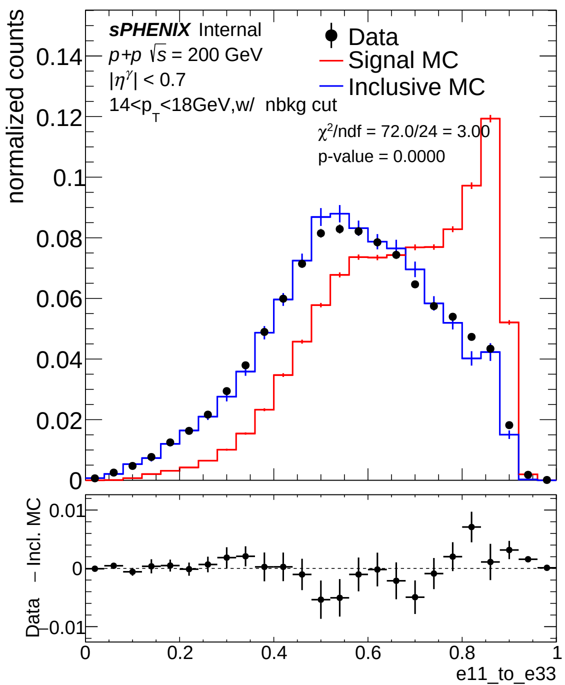
(a) f_1, Nominal MC



(b) f_1, Mixed MC (22.4% DI)



(c) $E_{1\times1}/E_{3\times3}$, Nominal MC

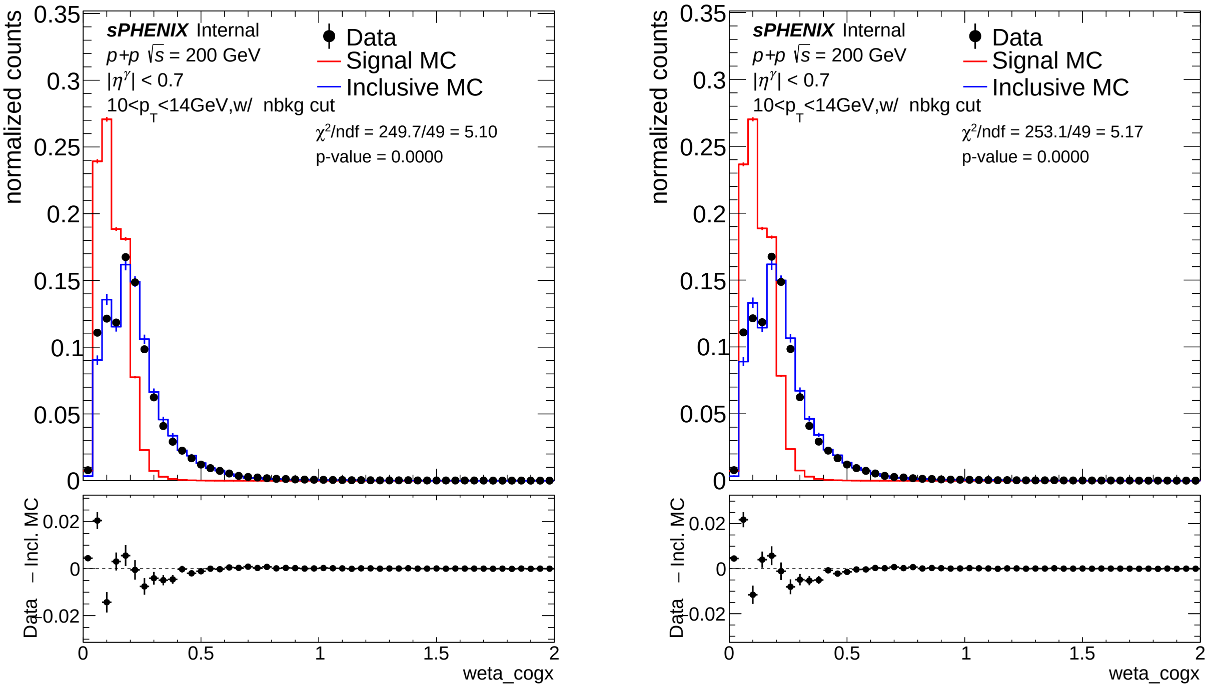


(d) $E_{1\times1}/E_{3\times3}$, Mixed MC (22.4% DI)

Figure 7: f_1 (top) and $E_{1\times1}/E_{3\times3}$ (bottom) at $14 < E_T < 18$ GeV (pt1), 0 mrad. f_1: $\chi^2/\text{ndf} = 5.65 \to 2.83$ (2.0×). $E_{1\times1}/E_{3\times3}$: $\chi^2/\text{ndf} = 4.74 \to 4.54$ (1.04×).

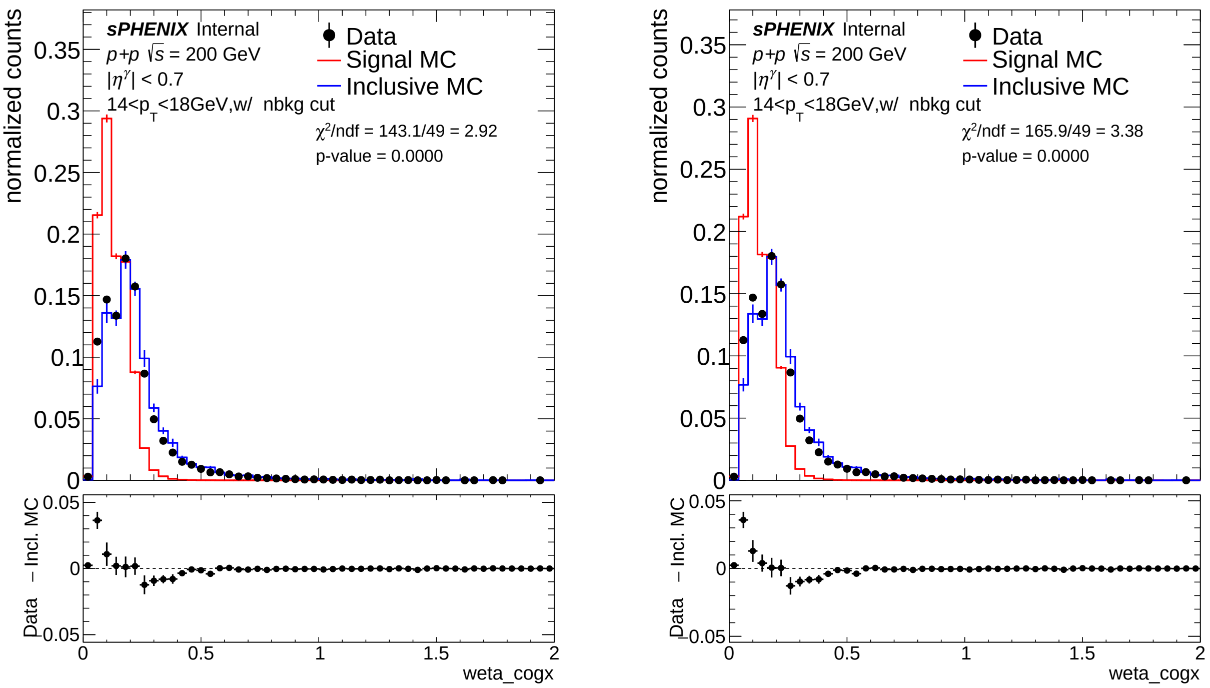
3 1.5 mrad Crossing Angle (7.9% Double Interaction)

3.1 w_η^{CoG} (Eta Cluster Width)



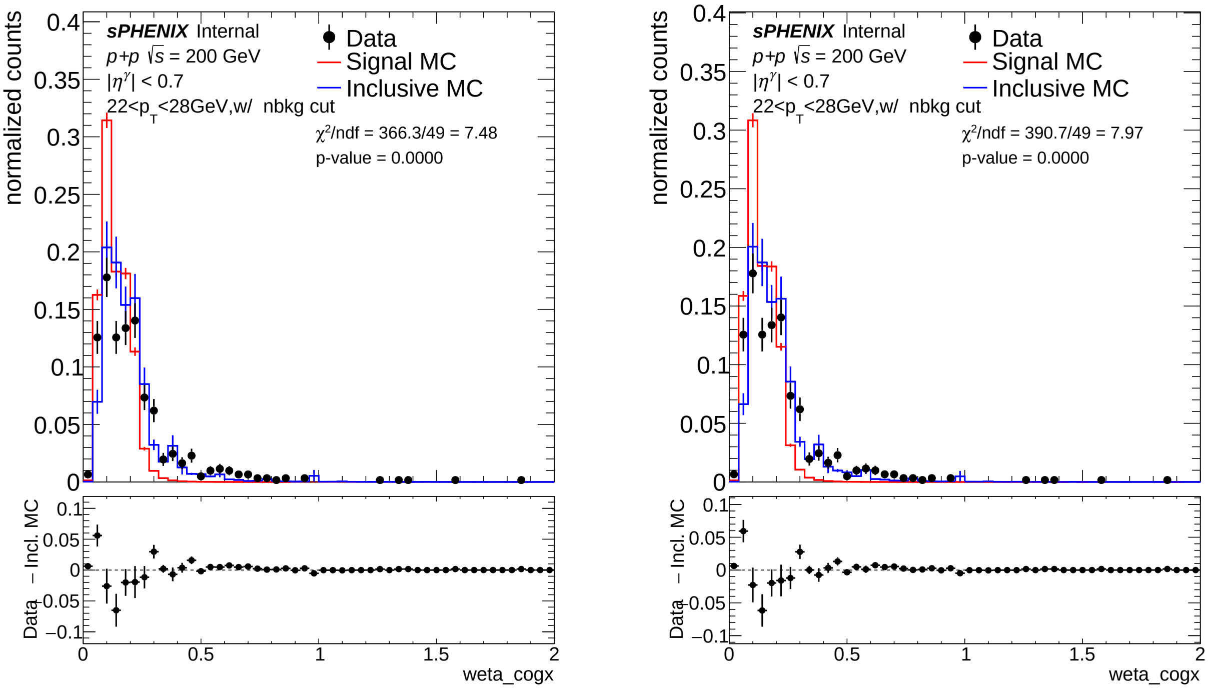
(a) Nominal MC (b) Mixed MC (7.9% DI)

Figure 8: w_η^{CoG}, $10 < E_T < 14$ GeV (pt0), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 5.20$; mixed $\chi^2/\mathrm{ndf} = 5.26$ (essentially flat, 1.01×).



(a) Nominal MC (b) Mixed MC (7.9% DI)

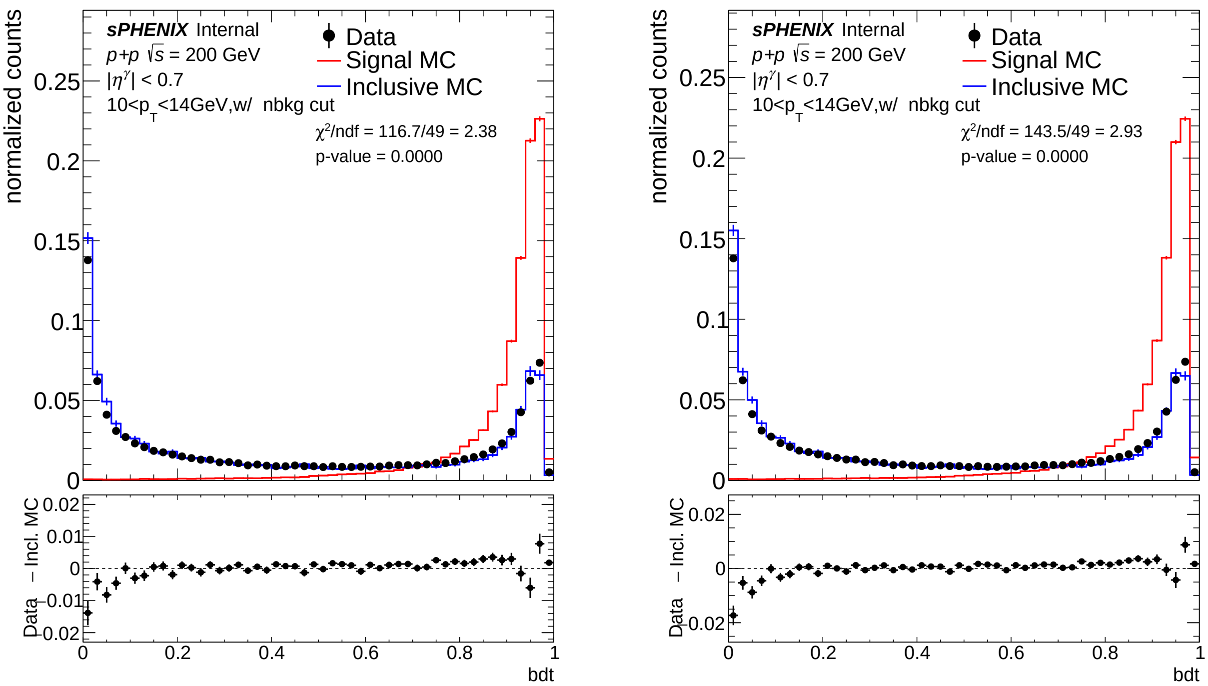
Figure 9: w_η^{CoG}, $14 < E_T < 18$ GeV (pt1), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 2.87$; mixed $\chi^2/\mathrm{ndf} = 3.32$ (1.16× **degradation**).



(a) Nominal MC (b) Mixed MC (7.9% DI)

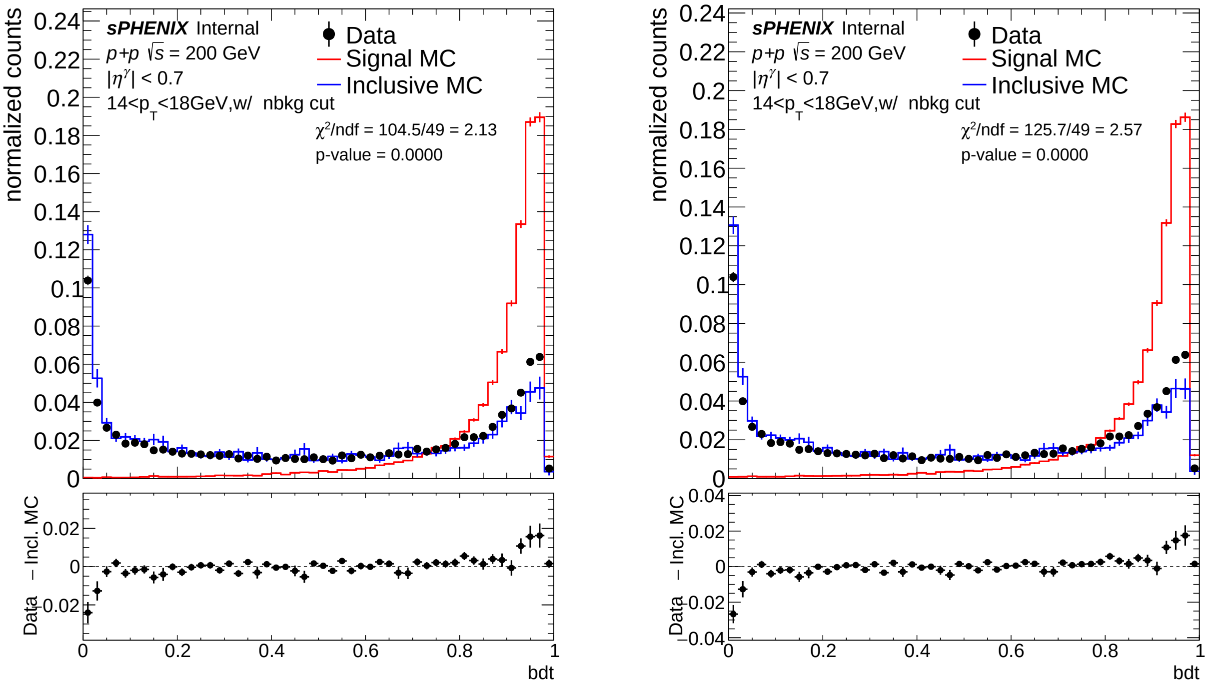
Figure 10: w_η^{CoG}, $22 < E_T < 28$ GeV (pt3), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 7.17$; mixed $\chi^2/\mathrm{ndf} = 7.61$ (1.06× **degradation**).

3.2 BDT Score



(a) Nominal MC (b) Mixed MC (7.9% DI)

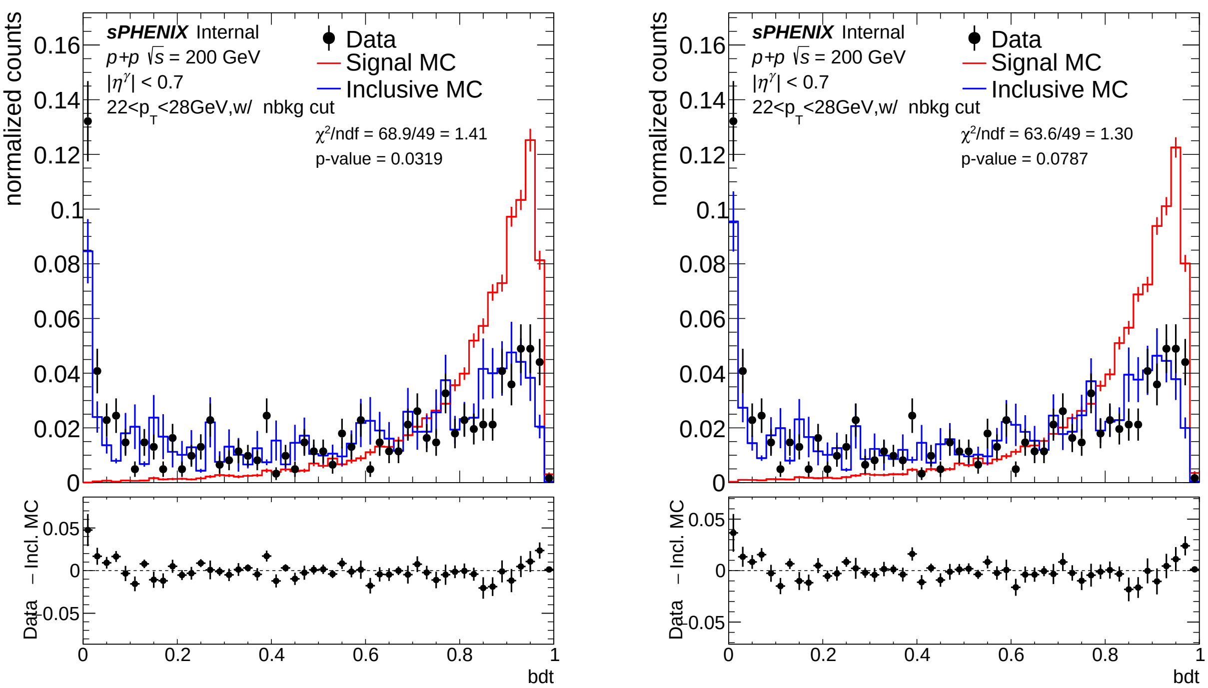
Figure 11: BDT score, $10 < E_T < 14$ GeV (pt0), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 2.37$; mixed $\chi^2/\mathrm{ndf} = 2.92$ (1.23× **degradation**).



(a) Nominal MC

(b) Mixed MC (7.9% DI)

Figure 12: BDT score, $14 < E_T < 18$ GeV (pt1), 1.5 mrad. Nominal χ^2/ndf = 2.13; mixed χ^2/ndf = 2.56 (1.20× **degradation**).

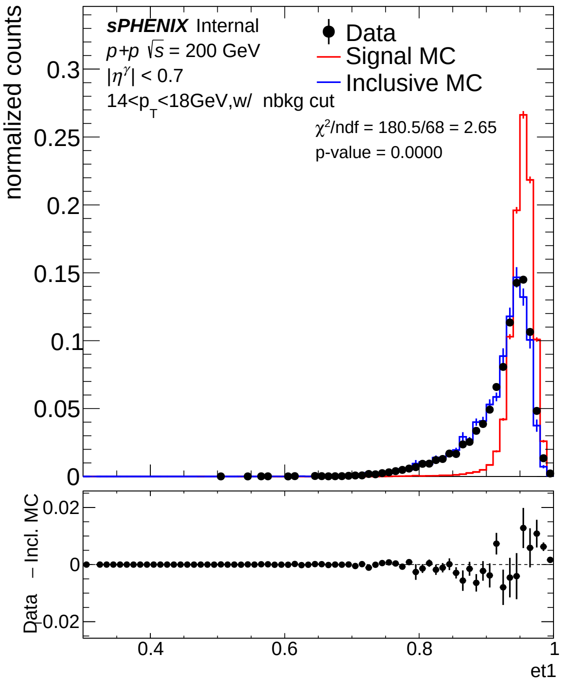


(a) Nominal MC

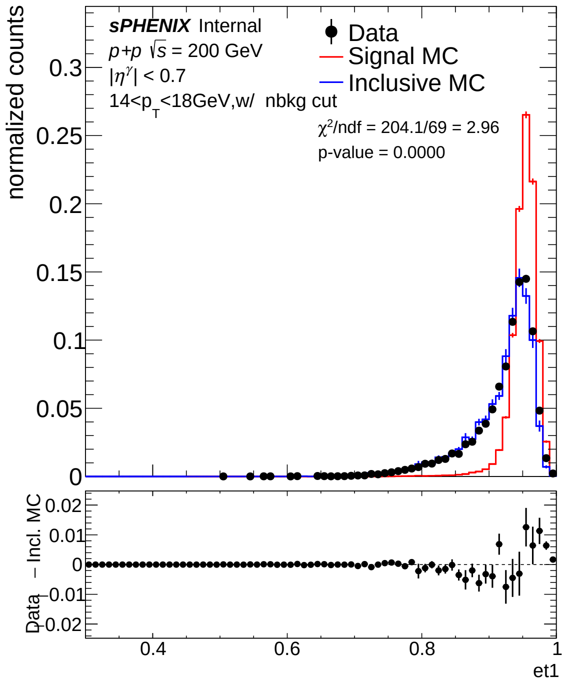
(b) Mixed MC (7.9% DI)

Figure 13: BDT score, $22 < E_T < 28$ GeV (pt3), 1.5 mrad. Nominal χ^2/ndf = 1.37 ($p = 0.042$); mixed χ^2/ndf = 1.27 ($p = 0.099$; 1.1× improvement).

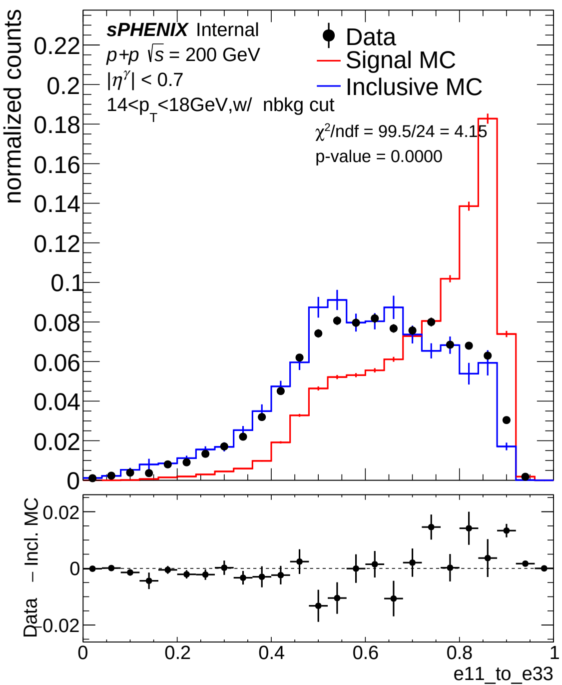
3.3 f_1 and $E_{1\times1}/E_{3\times3}$



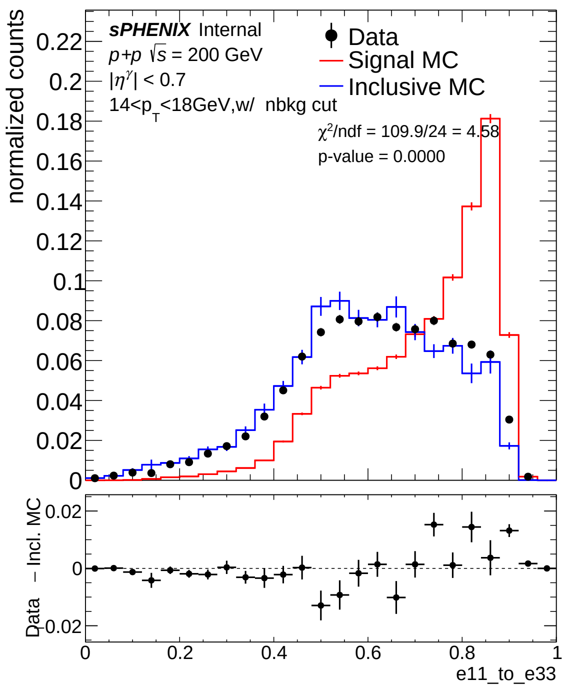
(a) f_1, Nominal MC



(b) f_1, Mixed MC (7.9% DI)



(c) $E_{1\times1}/E_{3\times3}$, Nominal MC

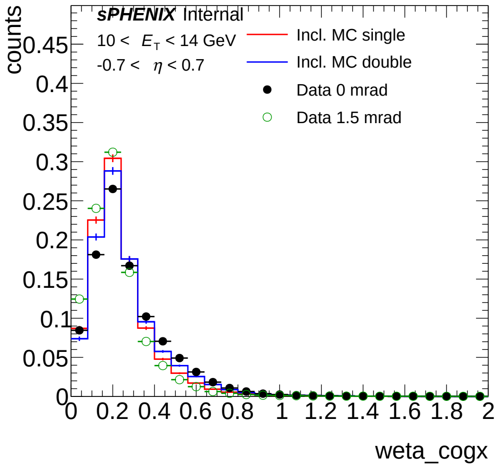


(d) $E_{1\times1}/E_{3\times3}$, Mixed MC (7.9% DI)

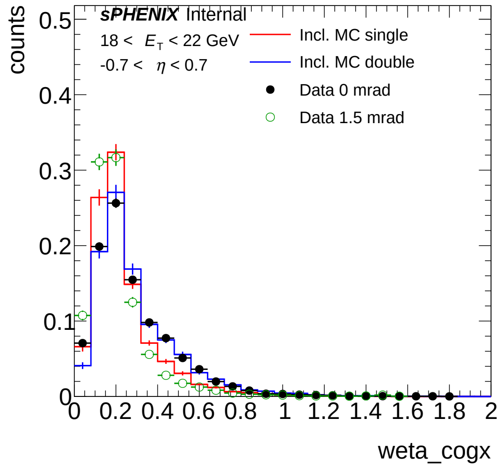
Figure 14: f_1 (top) and $E_{1\times1}/E_{3\times3}$ (bottom) at $14 < E_T < 18$ GeV (pt1), 1.5 mrad. f_1: $\chi^2/\mathrm{ndf} = 2.64 \to 2.94$ (1.11× degradation). $E_{1\times1}/E_{3\times3}$: $\chi^2/\mathrm{ndf} = 4.15 \to 4.59$ (1.11× degradation).

4 Single vs Double Interaction Overlay

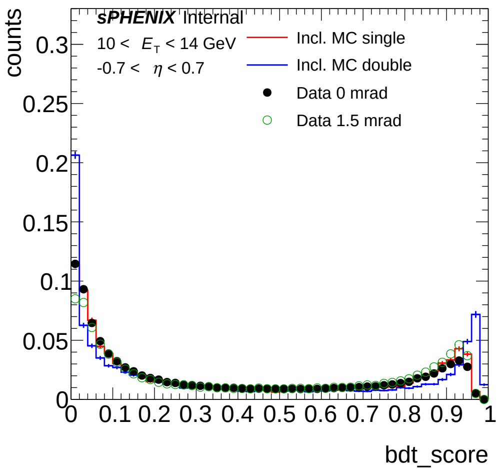
Figures 15 and 16 overlay single-interaction MC (red) and double-interaction MC (blue) with data from both crossing-angle periods: 0 mrad (filled black circles) and 1.5 mrad (open green circles), all area-normalized. Data at 0 mrad shows larger deviation from single MC, consistent with the higher pileup fraction, while data at 1.5 mrad tracks closer to single MC.



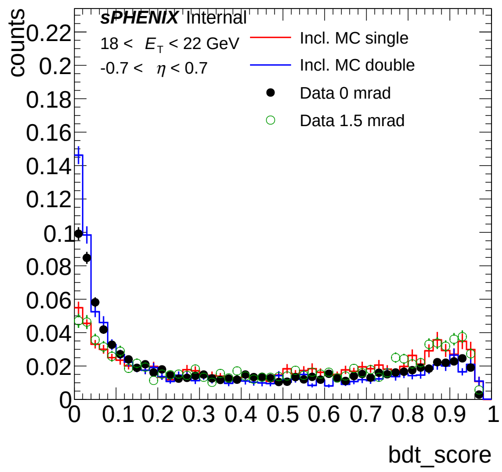
(a) w_η^{CoG}, 10–14 GeV



(b) w_η^{CoG}, 18–22 GeV

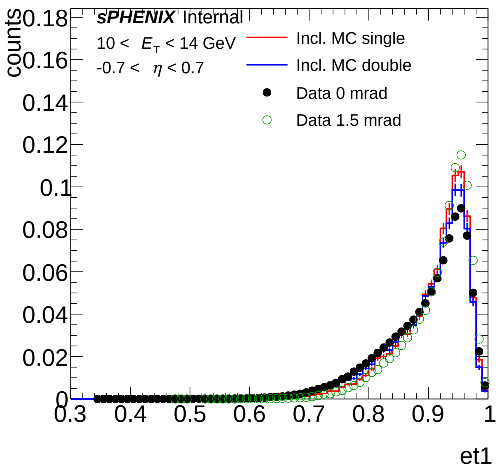


(c) BDT score, 10–14 GeV

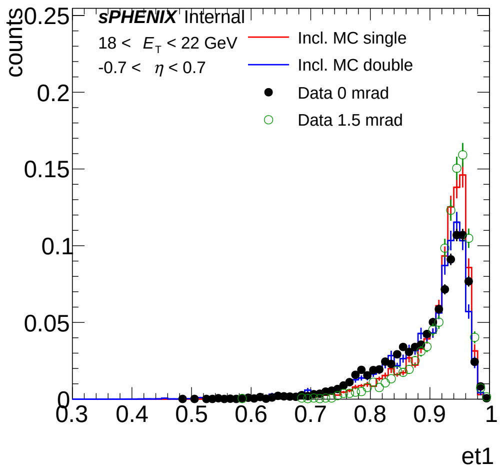


(d) BDT score, 18–22 GeV

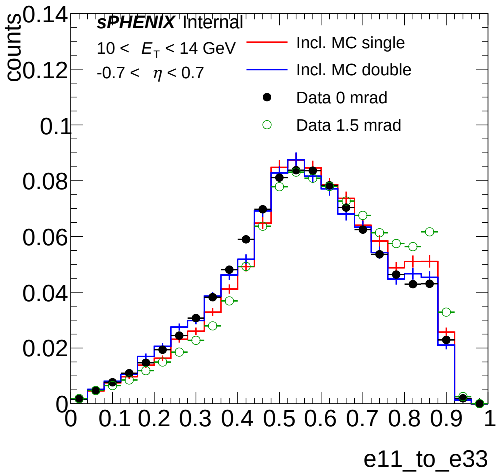
Figure 15: Shower shape distributions (part 1: w_η^{CoG} and BDT score) for single-interaction MC (red), double-interaction MC (blue), data 0 mrad (filled black circles), and data 1.5 mrad (open green circles), all area-normalized. Left column: $10 < E_T < 14$ GeV; right column: $18 < E_T < 22$ GeV.



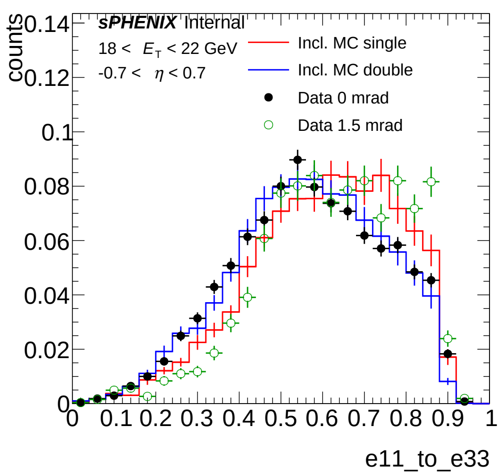
(a) f_1, 10–14 GeV



(b) f_1, 18–22 GeV



(c) $E_{1\times1}/E_{3\times3}$, 10–14 GeV



(d) $E_{1\times1}/E_{3\times3}$, 18–22 GeV

Figure 16: Shower shape distributions (part 2: f_1 and $E_{1\times1}/E_{3\times3}$) for single-interaction MC (red), double-interaction MC (blue), data 0 mrad (filled black circles), and data 1.5 mrad (open green circles), all area-normalized. Left column: $10 < E_T < 14$ GeV; right column: $18 < E_T < 22$ GeV.

5 χ^2/ndf Summary

Table 1 compiles the Data–Inclusive MC χ^2/ndf and p-values for all variables, E_T bins, and crossing angles. Entries where the mixed MC *degrades* agreement relative to nominal are highlighted in red. Improvement factors are computed as Nominal/Mixed when the mixed MC improves, and Mixed/Nominal (prefixed with a minus sign) when it degrades.

Table 1: Data–Inclusive MC χ^2/ndf and p-values at cut1 for nominal (single-only) and mixed (double-blended) MC. Lower χ^2/ndf is better; p-values $\ll 1$ indicate significant data–MC disagreement.

		0 mrad (22.4% DI)				**1.5 mrad (7.9% DI)**			
Variable	E_T [GeV]	Nom.	p	Mixed	p	Nom.	p	Mixed	p
w_η^{CoG}	10–14 (pt0)	28.27	<0.001	7.03	<0.001	5.20	<0.001	5.26	<0.001
w_η^{CoG}	14–18 (pt1)	7.39	<0.001	4.06	<0.001	2.87	<0.001	3.32	<0.001
w_η^{CoG}	22–28 (pt3)	7.02	<0.001	4.99	<0.001	7.17	<0.001	7.61	<0.001
BDT score	10–14 (pt0)	5.81	<0.001	2.84	<0.001	2.37	<0.001	2.92	<0.001
BDT score	14–18 (pt1)	6.97	<0.001	2.59	<0.001	2.13	<0.001	2.56	<0.001
BDT score	22–28 (pt3)	4.42	<0.001	2.01	<0.001	1.37	0.042	1.27	0.099
f_1	14–18 (pt1)	5.65	<0.001	2.83	<0.001	2.64	<0.001	2.94	<0.001
$E_{1\times1}/E_{3\times3}$	14–18 (pt1)	4.74	<0.001	4.54	<0.001	4.15	<0.001	4.59	<0.001

6 Discussion

6.1 Universal pattern: E_T-dependent crossover

A consistent pattern emerges across *both* crossing angles: the double-interaction blending **degrades** data–MC agreement for the BDT score at low E_T (pt0, 10–14 GeV) while **improving** it at higher E_T. This is now confirmed for both 0 mrad (3.0× BDT degradation at pt0) and 1.5 mrad (5.5× BDT degradation at pt0). For w_η^{CoG}, the crossover occurs between pt0 and pt1: at 0 mrad, the pt0 w_η^{CoG} still improves (1.6×), but at 1.5 mrad it degrades (4.4×), indicating that the crossover pileup fraction for w_η^{CoG} lies between 7.9% and 22.4% at low E_T.

6.2 Crossing-angle comparison

The improvement factors scale with the pileup fraction: the 22.4% admixture at 0 mrad yields larger improvements (up to 20.8× for w_η^{CoG} at pt3) than the 7.9% admixture at 1.5 mrad (up to 16.1×). Conversely, the 1.5 mrad nominals are already much better than their 0 mrad counterparts (e.g., BDT pt1: 1.16 vs. 12.20), confirming that lower pileup naturally reduces the data–MC discrepancy.

6.3 Low-E_T degradation mechanism

The low-E_T degradation affects the BDT score most severely because the BDT amplifies subtle shower shape distortions through its nonlinear decision function. Contributing factors include:

1. **Shared MC samples**: the same `photon10_double` and `jet12_double` samples are used for both crossing angles; beam conditions in the MC may better match one period than the other.
2. **Low-E_T sensitivity**: clusters near the 8–14 GeV threshold are most sensitive to vertex shifts and energy redistribution.
3. **Small-fraction noise**: at 7.9% mixing, any mismatch between the double MC and real pileup gets amplified relative to the small physical correction.

7 Conclusions and Recommendations

1. **0 mrad period**: the 22.4% double-interaction blending substantially improves data–MC agreement for pileup-sensitive variables (w_η^{CoG}, BDT, f_1) at $E_T \geq 14$ GeV, with improvement factors up to 20.8×. At pt0 (10–14 GeV), the BDT score degrades by 3.0× while w_η^{CoG} still improves modestly.

2. **1.5 mrad period**: the 7.9% blending helps at high E_T ($\geq 18\,\mathrm{GeV}$) but degrades both w_η^{CoG} and BDT at low E_T. The nominal agreement is already acceptable ($\chi^2/\mathrm{ndf} \lesssim 10$ for most variables).

3. **Recommended baseline for cross-section measurement**:

   - **0 mrad, $E_T \geq 14$ GeV**: use mixed MC (22.4% DI)
   - **0 mrad, $E_T < 14$ GeV**: use mixed MC for w_η^{CoG}; use nominal MC for BDT-sensitive quantities; assign nominal–mixed difference as systematic
   - **1.5 mrad, $E_T \geq 18$ GeV**: use mixed MC (7.9% DI)
   - **1.5 mrad, $E_T < 18$ GeV**: use nominal MC; assign nominal–mixed difference as systematic

4. **Alternative**: use nominal MC for the full 1.5 mrad period and apply the double-interaction correction to 0 mrad data only.